High-Level Architecture

At a high level, Dynamo is a Distributed Hash Table (DHT) that is replicated across the cluster for high availability and fault tolerance.

Introduction: Dynamo's architecture

Dynamo's architecture can be summarized as follows (we will discuss all of these concepts in detail in the following lessons):

1. Data distribution

Dynamo uses **Consistent Hashing** to distribute its data among nodes. Consistent hashing also makes it easy to add or remove nodes from a Dynamo cluster.

2. Data replication and consistency

Data is replicated optimistically, i.e., Dynamo provides eventual consistency.

3. Handling temporary failures

To handle temporary failures, Dynamo replicates data to a **sloppy quorum** of other nodes in the system instead of a strict majority quorum.

4. Inter-node communication and failure detection

Dynamo's nodes use **gossip protocol** to keep track of the cluster state.

5. High availability

Dynamo makes the system "always writeable" (*or highly available*) by using **hinted handoff.**

6. Conflict resolution and handling permanent failures

Since there are no write-time guarantees that nodes agree on values, Dynamo resolves potential conflicts using other mechanisms:

- Use **vector clocks** to keep track of value history and reconcile divergent histories at read time.
- In the background, dynamo uses an anti-entropy mechanism like **Merkle trees** to handle permanent failures.

Let's discuss each of these concepts one by one.